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Original Contribution

Gallbladder Disease and Risk of Type 2 Diabetes in Postmenopausal Women: A Women's Health Initiative Study

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Studies have suggested that adults with gallbladder disease have increased risk of type 2 diabetes. This prospective cohort study assessed the risk of type 2 diabetes in postmenopausal women with gallbladder disease. Data from women enrolled in the Women's Health Initiative from 1993 to 2005, aged 50–79 years (mean = 63.2; standard deviation, 7.2), were analyzed. Cox proportional hazards regression models were used to estimate the risk of type 2 diabetes associated with gallbladder disease. There were 8,896 new cases of type 2 diabetes after 1,025,486 person-years of follow-up. Gallbladder disease was significantly associated with type 2 diabetes (hazard ratio = 1.52; 95% confidence interval (CI): 1.38,1.67). The observed risk of type 2 diabetes in women with both gallbladder disease and central obesity was 37% higher than expected (relative excess risk due to interaction = 0.37, 95% CI: 0.11,0.63) on the additive scale. The hazard ratios for type 2 diabetes associated with gallbladder disease were 1.25 (95% CI: 1.19,1.32) and 1.48 (95% CI: 1.34,1.63) in women with and without central obesity, respectively, on the multiplicative scale. Results of this study support further studies to determine whether interventions in older women with gallbladder disease would reduce type 2 diabetes risk, especially among those with central obesity. Future research should examine the pathophysiological basis of the association between gallbladder disease and type 2 diabetes.

gallbladder disease; gut microbiome; hormone therapy; obesity; type 2 diabetes; women's health

Abbreviations: CI, confidence interval; HR, hazard ratio; WHI, Women's Health Initiative.

In the United States, an estimated 34.1 million adults had diabetes in 2018—13% of the adult population (1). In the same year, 1.5 million new cases of diabetes were diagnosed among adults, with higher incidence rates recorded among individuals aged 45 years or older than among those aged 18–44 years (1). The total cost (including direct medical expenditures and costs due to loss of productivity) of diagnosed diabetes was estimated to have climbed from $245 billion in 2012 to $327 billion in 2017 (2). Care for people with diabetes accounted for 1 out of every 4 dollars spent on health care in the United States (2).

About 20 to 25 million persons in the United States have gallstones—about 10% to 15% of the adult population (3, 4). Gallstones are more prevalent among women (4), and incidence increases with age, rising markedly after age 40 (3). Up to 80% of patients with gallstones remain asymptomatic and are likely to go undiagnosed (5). Symptomatic gallstones—called gallstone disease or gallbladder disease—present as episodes of abdominal pain (biliary colic) and can be complicated by life-threatening inflammation of the gallbladder, bile ducts, and pancreas (6, 7). Cholecystectomy, the mainstay of treatment for gallbladder disease, is one of the most commonly performed elective surgical procedures in the United States, with about 403,000 cases performed in 2007 (8).

While much of prior research has focused on the increased risk of gallstones among people with diabetes (9), emerging evidence suggests that this association is bidirectional. Recent prospective cohort studies have observed an increased risk of type 2 diabetes in persons with

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self-reported or ultrasound-diagnosed gallstones (10–12). Potential pathophysiological mechanisms linking gallbladder disease to type 2 diabetes include gallstone-associated chronic inflammation of the pancreas (13–15) and the alteration of gut microbial composition (in favor of pro-inflammatory bacterial species, for example) that may be seen in lithogenic states (16, 17).

Furthermore, while one prospective study (12) found no significant difference in the risk of type 2 diabetes in persons with gallstones who had had a cholecystectomy, compared with persons with gallstones without cholecystectomy, a cross-sectional study (18) found that metabolic syndrome—a component of which is impaired glucose tolerance (19)—was significantly associated with cholecystectomy but not gallstones without cholecystectomy. The effect of cholecystectomy on type 2 diabetes risk in persons with gallbladder disease has not been sufficiently investigated.

Prior research (10, 11) suggests interaction between gallbladder disease and central obesity on type 2 diabetes risk. The nature of this interaction remains to be fully clarified, however, as only multiplicative and not additive (biological interaction) was explored. Additionally, when men and women were studied separately, a significant association between gallstones and diabetes was found only in women (11, 12). In women, estrogen therapy has been linked to the formation of gallstones (20, 21) and to decreased incidence of type 2 diabetes (22–24); however, no published studies have, to our knowledge, examined the interaction between hormone therapy and gallbladder disease with respect to type 2 diabetes.

To explore previously unexamined associations and to further investigate the findings of previous studies, we used data from the Women's Health Initiative (WHI) clinical trial and observational study cohorts to examine the relationship between gallbladder disease and type 2 diabetes and assess potential multiplicative and additive interaction effects.

METHODS

Study population

The WHI is a large, prospective cohort of women aged 50–79 years at enrollment who were recruited at 40 clinical centers across the United States between October 1, 1993, and December 31, 1998. Participants were enrolled into one of 2 cohorts—the clinical trial (WHI-CT) and the observational study (WHI-OS). The clinical trial had 3 overlapping components, each a randomized double-blind clinical trial: a hormone therapy trial (2 arms, estrogen-only therapy and estrogen plus progesterone therapy), a dietary modification trial, and the calcium with vitamin D supplementation trial. In the estrogen-only arm of the hormone therapy trial, 10,739 women were randomized 1:1 to receive either 0.625 mg/day of conjugate equine estrogen or placebo, while in the estrogen-plus-progesterone arm, 16,608 women were randomized 1:1 to receive either 0.625 mg/day of estrogen plus 2.5 mg/day of medroxyprogesterone or placebo. The observational study enrolled a total of 93,676 women.

At baseline, self-reported data on treated diabetes, physician-diagnosed gallbladder disease, and cholecystectomy were collected via questionnaire, as were other elements of the medical history, lifestyle, and behavioral factors. Height, weight, and waist circumference were measured at baseline. Medical history, including incident diagnosis of gallbladder disease and type 2 diabetes, were updated during follow-up for all study participants—annually by mail for observational study participants and during annual clinic visits and at intermediate 6-month intervals (by mail) for clinical trial participants. All women were followed until study close-out in 2005, while a subset of participants continued to be followed in WHI Extension studies from 2005 to 2010 and from 2010 to 2020. Details of the design and implementation of the WHI are presented elsewhere (25, 26). This analysis included data from 1993 until 2005, the years during which data on incident gallbladder disease were collected.

Of the 161,808 women enrolled in the WHI, 7,169 participants reported having treated diabetes at baseline and 146 were missing this information and were excluded from the study, for a total of 7,315. Furthermore, participants were excluded if they were missing baseline data on gallbladder disease ($n = 1{,}032$), reported a history of liver cancer or other liver disease or pancreatitis at baseline ($n = 4{,}597$), or were missing follow-up data on treated diabetes ($n = 801$) or missing data on any key covariate ($n = 11{,}189$). After implementing exclusion criteria, 136,874 women (84.6% of total participants in the cohort) remained for this longitudinal study (Figure 1).

Outcome assessment

Incident diabetes was defined by self-report of physician-diagnosed diabetes treated with prescription medication. These data, including the number of days from study enrollment to diagnosis, were collected during follow-up in all participants. The validity of self-report of incident diabetes in WHI has previously been assessed and found to be sufficiently accurate for use in epidemiologic studies (27). Concordance between self-reported treated diabetes and medication inventory was 79% and 77% in the clinical trial and observational study participants, respectively, with similar rates recorded during the first 3 years of follow-up. Similarly, there was a 99.99% concordance between nonreport of treated diabetes and absence of antidiabetic drugs in the medication inventory (27).

Exposure and covariate assessment

The primary exposure was gallbladder disease. Data on gallbladder disease was collected at baseline. Participants were asked, "Did a doctor ever say that you had gallbladder disease or gallstones?" "Did you ever have a procedure to remove gallstones?" and "Did you have your gallbladder removed?" During follow-up, incident physician-diagnosed gallbladder disease continued to be reported by participants until study close-out in 2005. Data on cholecystectomy was collected at baseline but not during follow-up.

Potential confounders included as covariates in the analysis were identified a priori. Key covariates were assessed

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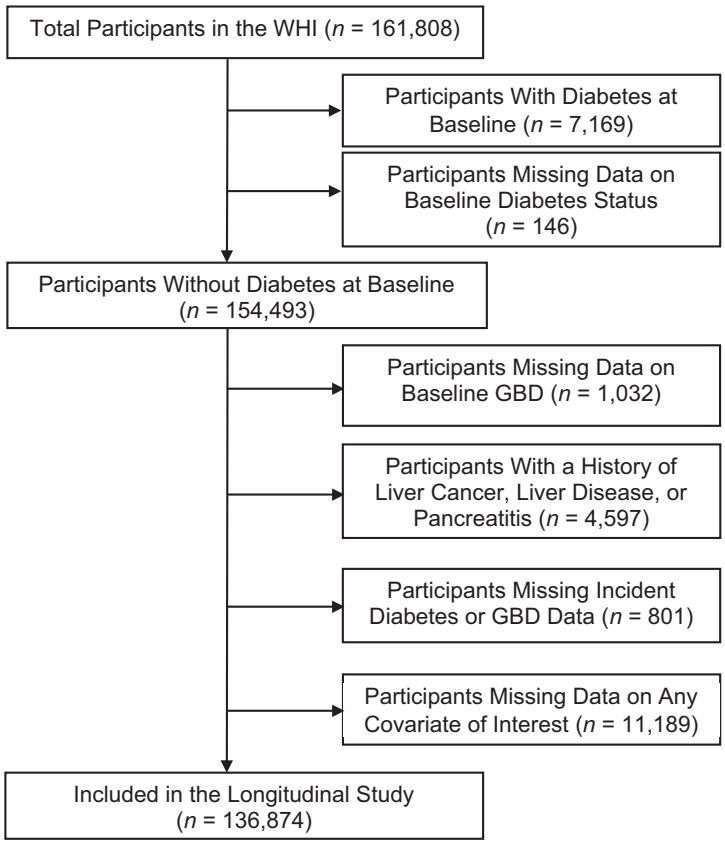

Figure 1. Flow diagram for participants included in a study of gallbladder disease (GBD) and type 2 diabetes, Women's Health Initiative (WHI), 1993–2005.

at baseline, including age, self-identified race/ethnicity, education, alcohol intake, smoking, physical activity, family history of diabetes, Healthy Eating Index-2010 (HEI) adherence, waist circumference/central obesity, postmenopausal hormone use, and the study component in which a participant was enrolled. Physical activity was measured as the total energy expended in recreational physical activity per week in metabolic equivalent of task (MET) hours (28). Waist circumference was used as a dichotomous variable, with a measurement greater than 35 inches indicating central obesity (19, 29). Waist circumference is commonly used as a measure for obesity across guidelines, including the National Cholesterol Education Program Adult Treatment Panel III, National Institutes of Health, European Group for the Study of Insulin Resistance, and the International Diabetes Foundation (19). The Healthy Eating Index-2010 is a measure of diet quality that was developed using key recommendations of the *Dietary Guidelines for Americans, 2010,* 7th Edition (30, 31). Diet components measured in the HEI-2010 include vegetables and fruit, grains, dairy, protein foods, fats, sodium, and alcohol (31). Previous studies have shown an association between HEI-2010 scores and type 2 diabetes (32, 33). For participants in the hormone therapy trial, postmenopausal hormone therapy was indicated by randomization to the experimental group and for those not in the hormone therapy trial, postmenopausal hormone therapy was indicated by self-report of ongoing use of prescribed estrogen or combined estrogen and progesterone at enrollment.

Statistical analysis

We used Student's unpaired *t* tests and χ^2 tests to compare characteristics of study participants according to gallbladder disease status at baseline. We also compared the characteristics of WHI participants excluded from the study with the included participants, looking for any differences that may suggest selection bias. We estimated the association between gallbladder disease and type 2 diabetes using multivariable Cox proportional hazards models. Entry time was defined as day of study enrollment while exit time was defined as time of diagnosis of treated diabetes, loss to follow-up, death, or end of study follow-up. Gallbladder disease was modeled as a time-varying covariate, with status changing during follow-up from "no gallbladder disease" to "gallbladder disease" at the point of diagnosis. Interaction between gallbladder disease and central obesity was tested on additive and multiplicative scales to provide complementary information. The interaction on the additive scale

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describes the excess risk as a result of the combined versus individual risk factors. Additive interaction was assessed by calculating the relative excess risk due to interaction (RERI) from terms in the standard Cox proportional hazards model using the delta method (33, 34). Interaction between gallbladder disease and postmenopausal hormone therapy was similarly tested. Using baseline exposure—since data on cholecystectomy were collected only at enrollment—the risks of type 2 diabetes associated with gallbladder disease treated with cholecystectomy and gallbladder disease without cholecystectomy were estimated, with the comparison group as women with no disease. Participants were censored when they developed gallbladder disease since cholecystectomy status was unknown beyond this point. Proportional hazards assumption was tested using graphical diagnostics based on scaled Schoenfield residuals and seen to have been satisfied by all models.

We carried out sensitivity analyses. First, we tested for reverse causation by repeating the primary analysis after excluding participants who developed type 2 diabetes within a year of study enrollment (for participants positive for gallbladder disease at baseline) or within a year of gallbladder disease diagnosis. Second, to see if results remain unchanged when a measure of obesity besides waist circumference was used, we repeated the analysis after dropping participants with a body mass index (weight (kg)/height $(m)^2$) above 30 at enrollment. Third, to ensure that model estimates were not significantly affected by loss of information due to dichotomizing waist circumference into "central obesity" and "no central obesity," we modeled waist circumference as a continuous variable and compared the resulting model with the primary analysis. Fourth, we excluded participants who developed gallbladder cancer, pancreatic cancer, pancreatitis, liver cancer, or other liver disease during follow-up to test the possibility of confounding or mediation of the association of interest by these factors. Two-sided tests of hypotheses, with α level of significance set a priori to 0.05, were performed throughout. Statistical analyses were performed using R, version 3.6.1 (R Foundation for Statistical Computing, Vienna, Austria) (35).

RESULTS

Of the women in the study sample, 82,100 (60%) were enrolled in the WHI observational study while 54,774 (40%) were enrolled in at least 1 of the 3 WHI randomized trials. The mean age of participants at baseline was 63.2 (standard deviation, 7.2) years. The study population was racially and ethnically diverse; it included women who were Non-Hispanic Black (8.2%), Non-Hispanic White (83.9%), Hispanic or Latino (3.7%), and American Indian, Alaskan Native, Asian, or Pacific Islander (4.2%). At baseline, 116,573 (85.2%) of women had completed high school and 51,395 (37.5%) had waist circumference greater than 35 inches, indicating central obesity. A total of 20,892 (15.3%) women had already been diagnosed with gallbladder disease at baseline. Compared with women without gallbladder disease, women with gallbladder disease at baseline had fewer years of education and lower physical activity levels and were more likely to have central obesity. Information on the baseline characteristics of the study population is presented in Table 1. Participants excluded from the final study population did not differ appreciably from those included (Web Table 1, available at https://doi.org/10.1093/aje/kwac074).

There were 8,896 new cases of type 2 diabetes after 1,025,486 person-years of follow-up (8.67 cases per 1,000 person-years) in the entire cohort. The mean duration between diagnosis of gallbladder disease and diagnosis of type 2 diabetes was 3.32 (standard deviation, 1.86) years. On average, participants who did not develop diabetes were followed for 8.5 years. After adjusting for age, race/ethnicity, education, physical activity, family history of diabetes, smoking, central obesity, Healthy Eating Index adherence, and study component in which the participant was enrolled, gallbladder disease was significantly associated with increased risk of type 2 diabetes (hazard ratio (HR) = 1.52, 95% confidence interval (CI): 1.38, 1.67) (Table 2).

We observed significant positive interaction on the additive scale; the relative excess risk due to interaction was 0.37 (95% CI: 0.11, 0.63), meaning that the observed combined effect of gallbladder disease and central obesity was 37% more than the expected sum of their individual effects. This indicates a higher-than-expected risk of type 2 diabetes in women with gallbladder disease who also had central obesity, as compared with women with gallbladder disease without central obesity. In contrast, the observed interaction on the multiplicative scale was negative. Stratified by central obesity status, estimated HR for type 2 diabetes associated with gallbladder disease in the obesity group was 1.25 (95% CI: 1.19, 1.32) compared with 1.48 (95% CI: 1.34, 1.63) in the non–obesity group (Table 3).

While postmenopausal hormone therapy was significantly associated with decreased risk of type 2 diabetes (HR = 0.88, 95% CI: 0.85, 0.92), there was no evidence of interaction with gallbladder disease (P = 0.84). The full model is presented in Web Table 2. When compared with women without gallbladder disease, the HRs for type 2 diabetes in those with gallbladder disease treated with cholecystectomy and those who had gallbladder disease but not cholecystectomy were 1.58 (95% CI: 1.40, 1.77) and 1.31 (95% CI: 1.06, 1.61), respectively (Table 2).

In sensitivity analyses to test for reverse causation (i.e., whether the observed effect is explained by an increased risk of gallbladder disease due to diabetes or prediabetes), excluding participants diagnosed with type 2 diabetes within a year of gallbladder disease diagnosis did not eliminate the risk of type 2 diabetes associated with gallbladder disease (HR = 1.48, 95% CI: 1.34, 1.63). Restricting the analysis to participants with a body mass index of 30 or below did not attenuate or eliminate the observed association (HR= 1.46, 95% CI: 1.35, 1.58). Using waist circumference as a continuous variable in the model (as opposed to dichotomous) produced similar risk estimates for type 2 diabetes associated with gallbladder disease (HR = 1.34, 95% CI: 1.26, 1.42). Excluding participants who developed gallbladder cancer, pancreatic cancer, pancreatitis, liver cancer, or other liver disease during follow-up also produced similar results (HR = 1.50, 95% CI: 1.36, 1.65).

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Table 1. Baseline Characteristics of Participants in an Analysis of Gallbladder Disease and Type 2 Diabetes, Women's Health Initiative, 1993–1998

Characteristic	Total (*n* = 136,874)		GBD (*n* = 20,892)		No GBD (*n* = 115,982)		*P* Value
	No.	%	No.	%	No.	%	
Age, years[a]	63.2 (7.2)		64.2 (7.1)		63.0 (7.2)		<0.0001
Race/ethnicity							<0.0001
Non-Hispanic White	114,818	83.9	17,988	86.1	96,830	83.5	
Non-Hispanic Black	11,272	8.2	1,157	5.5	10,115	8.7	
Hispanic/Latino	5,038	3.7	1,074	5.1	3,964	3.4	
Others[b]	5,746	4.2	673	3.2	5,073	4.4	
Education							<0.0001
Lower than high school or vocational	20,301	14.8	3,904	18.7	16,397	14.1	
High school or higher	116,573	85.2	16,988	81.3	99,585	85.9	
Alcohol intake							<0.0001
Nondrinker	14,380	10.5	2,505	12.0	11,875	10.2	
Past drinker	23,731	17.3	4,473	21.4	19,258	16.6	
<7 drinks per week	82,062	60.0	12,210	58.4	69,852	60.2	
≥7 drinks per week	16,701	12.2	1,704	8.2	14,997	12.9	
Smoking							0.07
Never smoked	69,633	50.9	10,522	50.4	59,111	51.0	
Past smoker	57,821	42.2	8,970	42.9	48,851	42.1	
Current smoker	9,420	6.9	1,400	6.7	8,020	6.9	
Physical activity, MET-hours/week							<0.0001
No activity	21,086	15.4	4,044	19.4	17,042	14.7	
<5.0	28,263	20.6	5,004	24.0	23,259	20.1	
5.0–9.9	23,679	17.3	3,756	18.0	19,923	17.2	
10.0–19.9	33,391	24.4	4,530	21.7	28,861	24.9	
≥20.0	30,455	22.3	3,558	17.0	26,897	23.2	
Waist circumference, inches[a]	33.8 (5.3)		36.0 (5.6)		33.4 (5.2)		<0.0001
Central obesity							<0.0001
No	85,479	62.5	9,478	45.4	76,001	65.5	
Yes	51,395	37.5	11,414	54.6	39,981	34.5	
First-degree relative with diabetes							<0.0001
No	88,534	64.7	12,568	60.2	75,966	65.5	
Yes	42,098	30.8	7,348	35.2	34,750	30.0	
Unknown	6,242	4.6	976	4.7	5,266	4.5	
HEI score[a]	65.3 (10.4)		63.6 (10.7)		65.6 (10.4)		<0.0001
Postmenopausal hormone therapy							<0.0001
No	70,402	51.4	10,490	50.2	59,912	51.7	
Yes	66,472	48.6	10,402	49.8	56,070	48.3	
WHI Observational Study							<0.001
No	54,774	40.0	8,582	41.1	46,192	39.8	
Yes	82,100	60.0	12,310	58.9	69,790	60.2	

Table continues

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Table 1. Continued

Characteristic	Total (*n* = 136,874)		GBD (*n* = 20,892)		No GBD (*n* = 115,982)		*P* Value
	No.	%	No.	%	No.	%	
WHI Hormone Trial							0.03
No	114,945	84.0	17,439	83.5	97,506	84.1	
Yes	21,929	16.0	3,453	16.5	18,476	15.9	
WHI Dietary Modification Trial							0.0005
No	97,805	71.5	14,719	70.5	83,086	71.6	
Yes	39,069	28.5	6,173	29.5	32,896	28.4	
Calcium and Vitamin D trial							0.45
No	107,243	78.4	16,411	78.6	90,832	78.3	
Yes	29,631	21.6	4,481	21.4	25,150	21.7	

Abbreviations: GBD, gallbladder disease; HEI, Health Eating Index; MET, metabolic equivalent of task; WHI, Women's Health Initiative.
[a] Values are expressed as mean (standard deviation).
[b] Includes American Indian or Alaskan Native and Asian or Pacific Islander.

DISCUSSION

We found that gallbladder disease was significantly associated with incident type 2 diabetes in postmenopausal women. To our knowledge, the present study is the first to look exclusively at a population of women. We observed significant interaction on both the additive and multiplicative scales between gallbladder disease and central obesity. Prior studies evaluated multiplicative but not additive interaction. It has been argued that testing for interaction on the additive scale is more relevant to clinical and public health decision-making as it allows a more accurate determination of target groups for intervention (36, 37).

As in previous studies (10, 11), we found significant negative interaction between gallbladder disease and central obesity on risk of type 2 diabetes on the multiplicative scale. In contrast, the significant interaction on the additive scale indicated higher-than-expected risk of type 2 diabetes in women with gallbladder disease who also had central obesity, as compared with women with gallbladder disease without central obesity. This scenario of negative multiplicative interaction but positive additive interaction occurs when the baseline risk for the outcome of interest is different in the subgroups defined by the interaction term (38) and has been previously observed in studies of smoking and asbestos exposure in lung cancer occurrence (39), *Helicobacter pylori* and nonsteroidal antiinflammatory drugs in peptic ulcer (40), and factor V Leiden mutation and oral contraceptive use in venous thrombosis (41). Illustrating this phenomenon in the current study, abdominal obesity is a well-established risk factor for type 2 diabetes (42, 43), and the difference in risk for type 2 diabetes associated with gallbladder disease was smaller than the difference in baseline risk comparing women with central obesity with those who did not have central obesity (see Web Table 2). Interaction on the additive scale is relevant for assessing biological

Table 2. Association of Type 2 Diabetes With Gallbladder Disease and Cholecystectomy, Women's Health Initiative, 1993–2005

Gallbladder Disease Status	Cases	Person Years of Follow-up	Model 1[a]		*P* Value	Model 2[b]		*P* Value
			HR	95% CI		HR	95% CI	
No GBD	6,378	842,852	1.00	Referent		1.00	Referent	
GBD	2,518	182,635	1.76	1.68, 1.84	<0.0001	1.52	1.38, 1.67	<0.0001
With cholecystectomy	1,675	118,988	1.86	1.76, 1.96	<0.0001	1.58	1.40, 1.77	<0.0001
Without cholecystectomy	453	32,836	1.81	1.64, 1.99	<0.0001	1.31	1.06, 1.61	0.02

Abbreviations: CI; confidence interval; GBD, gallbladder disease; HR, hazard ratio.
[a] Age-adjusted model.
[b] Adjusted for age, race/ethnicity, education, physical activity, family history, smoking, alcohol, central obesity, Healthy Eating Index score, and study component.

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Table 3. Association of Type 2 Diabetes With Gallbladder Disease, Stratified by Central Obesity, Women's Health Initiative, 1993–2005

	Cases	Person Years of Follow-up	Model 1[a]		*P* Value	Model 2[b]		*P* Value
			HR	95% CI		HR	95% CI	
Central obesity								
No GBD	4,299	275,673	1.00	Referent		1.00	Referent	
GBD	1,976	95,517	1.30	1.23, 1.37	<0.0001	1.25	1.19, 1.32	<0.0001
No central obesity								
No GBD	2,079	567,179	1.00	Referent		1.00	Referent	
GBD	542	87,119	1.55	1.41, 1.71	<0.0001	1.48	1.34, 1.63	<0.0001

Abbreviations: CI, confidence interval; GBD, gallbladder disease; HR, hazard ratio.
[a] Age-adjusted model.
[b] Adjusted for age, race/ethnicity, education, physical activity, family history, smoking, alcohol, Healthy Eating Index score, and study component.

interaction, as Rothman (44) has shown and Ahlbom and Alfredsson (45) have further argued. Biological interaction is present when 2 factors are required in at least 1 pathway to a disease (45). While gallbladder disease and central obesity are independently associated with type 2 diabetes, results of this study suggest there is at least a pathway to type 2 diabetes that involves both factors. Taken together, our results provide support for identifying all older women with gallbladder disease for intervention to prevent type 2 diabetes, but emphasizing prevention in those who also have central obesity would be of even greater benefit in terms of the absolute number of cases prevented.

While our study was not designed to evaluate the biological mechanisms underlying the association, existing research suggests a role for gut microbes in the pathophysiological processes linking gallstones and type 2 diabetes. The amount and composition of bile acids secreted into the gut modulate gut microbial volume and community structure, which in turn have been linked with the derangement of metabolic pathways involved in glucose homeostasis and obesity (16, 46–48). Low bile acid levels entering the gastrointestinal tract have, for example, been associated with higher numbers of pro-inflammatory gut bacteria (17, 49). The lack of a significant difference in risk of type 2 diabetes between participants with a cholecystectomy and those with gallbladder disease but no cholecystectomy may suggest that underlying lithogenic processes, rather than cholecystectomy, are implicated in the increased risk of type 2 diabetes observed.

Gallbladder sludge, gallstones, and complications of cholecystectomy have also been linked to chronic pancreatitis (13–15, 50), which could further explain the increased risk of type 2 diabetes seen in persons with gallbladder disease (51). While obesity has traditionally been independently associated with gallbladder disease and type 2 diabetes, persistence of the association after controlling for central obesity and the similar results obtained after excluding participants with body mass index in the obesity range suggests that obesity alone does not explain the observed result.

A strength of our study is the longitudinal design with a large sample size and the use of time-varying gallbladder disease, which ensured that exposure status was updated appropriately for women who developed gallbladder disease during follow-up. We were also able to explore more fully the nature of the interaction between gallbladder disease and central obesity with respect to the risk of type 2 diabetes. Some limitations are noted, however. Exclusions due to missing data may have introduced selection bias, but as included participants were similar to excluded participants, potential bias is limited. Self-reported exposure and outcome data may have resulted in some misclassification. The validity of self-reported treated diabetes in the Women's Health Initiative has, however, been previously established to be sufficiently accurate for use in epidemiologic studies (27). Also, data on cholecystectomy were not collected during follow-up, which limited our ability to examine the effect of cholecystectomy on the risk of type 2 diabetes. This remains an interesting topic of future research. For all participants, gallbladder disease was self-reported in this study, as opposed to by record of diagnosis by ultrasound, which would have detected even asymptomatic cases. As such, it is to be expected that most cases of gallbladder disease in this study represented symptomatic cases, an assumption that limits generalizing the results to include cases of asymptomatic gallstones.

Our study among postmenopausal women adds to the growing body of evidence that gallbladder disease is a significant risk factor for type 2 diabetes. In addition, we found evidence for a significant synergistic interaction between gallbladder disease and central obesity regarding risk of type 2 diabetes. These findings support future studies to determine whether interventions in women diagnosed with gallbladder disease would be effective in reducing the risk of type 2 diabetes given the evidence of higher risk in this group. The findings also suggest that interventions should especially target women who are also obese, given the evidence of particularly high risk in this group. Furthermore, studies examining the pathophysiological mechanisms underlying the association between gallbladder

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disease and type 2 diabetes will advance our understanding and possibly inform the design of preventive interventions.

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